Week 3 Reading

Weather and climate extreme events in a changing climate

1 Assigned Reading

> **Tip**
>
> As before, you are encouraged to use Zotero to manage your references. If you're adding this to Zotero, you may find the IPCC-Bibtex project helpful.

This week's reading is S. Seneviratne *et al.* [1], the IPCC AR6 chapter on "Weather and Climate Extreme Events in a Changing Climate."

1.1 Part 1: Background (Everyone)

Read these sections to understand the chapter's methodology:

- **Executive Summary** (pp. 1517-1522): Overview of key findings and confidence levels
- **Section 11.1 Introduction** (pp. 1523-1527): Framing and scope
- **Section 11.2 Data and Methods** (pp. 1527-1536): How observations and models are used to study extremes

1.2 Part 2: Deep Dive (Pick ONE Hazard)

Choose **one** of the following subsections and read it carefully. You will be asked to summarize it in class (and is fair game for the quiz).

Section	Hazard Type
11.3	Temperature extremes (heat waves, cold spells)
11.4	Heavy precipitation and pluvial floods
11.5	Floods (fluvial and coastal)
11.6	Droughts
11.7	Extreme storms (tropical cyclones, extratropical storms)
11.8	Compound events

1.3 Part 3: Class Contribution

Prepare a **two-minute verbal summary** of your deep-dive subsection to share during discussion. Focus on: What extremes are covered? How are they changing? What surprised you?

2 Discussion questions

> **Tip**
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> These questions are intended to guide your reading. You don't need to turn in answers, but are encouraged to write answers and bring them to class to discuss.

1. Share one climate extreme finding from your subsection that surprised you, and one that was less concerning than you expected. What drives the difference between high and low confidence statements in the IPCC report?
2. Section 11.2 discusses using observations and models to understand extremes. What are the strengths and limitations of each approach?
3. How does confidence level vary across hazard types in the different subsections (11.3-11.8)? Why might some extremes be easier to project than others?
4. If advising a city on flood infrastructure, how would you translate IPCC confidence statements into actionable guidance?

Bibliography

[1] S. Seneviratne *et al.*, "Weather and Climate Extreme Events in a Changing Climate," *Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*. Cambridge University Press, Cambridge, UK and New York, NY, USA, 2021. doi: 10.1017/9781009157896.013.